

Math 242, S13
Quiz 12

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (4 points) Use Theorem of Differential Transforms (i.e., $\mathcal{L}\{tf(t)\} = -F'(s)$ where $F(s) = \mathcal{L}\{f(t)\}$) to compute $\mathcal{L}\{t \sin 4t\}$.

Solution: Let $f(t) = \sin 4t$, then $F(s) = \mathcal{L}\{\sin 4t\} = \frac{4}{s^2 + 16}$

$$\begin{aligned}\Rightarrow \mathcal{L}\{t \sin 4t\} &= -F'(s) \\ &= -\frac{d}{ds} \left[\frac{4}{s^2 + 16} \right] \\ &= \frac{8s}{(s^2 + 16)^2}\end{aligned}$$

Problem 2. (6 points) Use Laplace transform to solve the initial value problem

$$x'' + 9x = 4t; \quad x(0) = x'(0) = 0.$$

Solution: Let $X(s) = \mathcal{L}\{x(t)\}$, then

$$\mathcal{L}\{x'(t)\} = sX(s)$$

$$\mathcal{L}\{x''(t)\} = s^2 X(s)$$

Thus

$$\mathcal{L}\{x'' + 9x\} = \mathcal{L}\{4t\}$$

$$\mathcal{L}\{x''\} + 9\mathcal{L}\{x\} = 4 \cdot \frac{1}{s^2}$$

$$s^2 X(s) + 9X(s) = \frac{4}{s^2}$$

$$(s^2 + 9)X(s) = \frac{4}{s^2}$$

$$X(s) = \frac{4}{s^2(s^2 + 9)}$$

$$\begin{aligned}\frac{4}{s^2(s^2 + 9)} &= \frac{A}{s^2} + \frac{B}{s^2 + 9} \\ &= \frac{A(s^2 + 9) + Bs^2}{s^2(s^2 + 9)} \\ &= \frac{(A+B)s^2 + 9A}{s^2(s^2 + 9)}\end{aligned}$$

$$\Rightarrow \begin{cases} A+B=0 \\ 9A=4 \end{cases} \Rightarrow \begin{aligned} A &= \frac{4}{9} \\ B &= -\frac{4}{9} \end{aligned}$$

$$\begin{aligned}\Rightarrow x(t) &= \mathcal{L}^{-1}\left\{ \frac{4}{s^2(s^2 + 9)} \right\} \\ &= \mathcal{L}^{-1}\left\{ \frac{4}{9} \frac{1}{s^2} - \frac{4}{9} \frac{1}{s^2 + 9} \right\} \\ &= \frac{4}{9}t - \frac{4}{27} \sin 3t\end{aligned}$$